50.5: The physical interaction of protein filaments is required for muscle function

In discussing sensory mechanisms, we have seen how sensory inputs to the nervous system can result in specific behaviors: the touch-guided foraging of a star-nosed mole, the upside-down swimming of a crayfish with manipulated statocysts, and the light-avoiding maneuvers of planarians. Underlying these diverse behaviors are common fundamental mechanisms: Feeding, swimming, and crawling all require muscle activity in response to nervous system motor output.

Muscle cell contraction relies on the interaction between protein structures called thin and thick filaments. The major component of **thin filaments** is the globular protein actin. In thin filaments, two strands of polymerized actin are coiled around one another; similar actin structures called microfilaments function in cell motility. The **thick filaments** are staggered arrays of myosin molecules. Muscle contraction is the result of filament movement powered by chemical energy; muscle extension occurs only passively. To understand how filaments bring about muscle contraction, we will begin by examining the skeletal muscle of vertebrates.

Vertebrate Skeletal Muscle

Vertebrate **skeletal muscle**, which moves bones and body, has a hierarchy of smaller and smaller units **(Figure 50.26)**. Within a typical skeletal muscle is a bundle of long fibers running along the length of the muscle. Each individual fiber is a single cell. Within are multiple nuclei, each derived from one of the stem cells that fused to form the fiber. Surrounding these nuclei are longitudinal **myofibrils**, which consist of bundles of thin and thick filaments.

Figure 50.26

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The structure of skeletal muscle.
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Structure of skeletal muscle includes bundles of muscle fibers, myofibril, and sarcomeres with thick and thin filaments.

Muscle (example bicep) consists of bundles of muscle fibers. Each muscle fiber is a single cell with multiple nuclei. Inside the muscle cell are bundles of myofibril (contain thick and thin filaments). Myofibrils are made up of repeating segments called sarcomeres bordered by Z lines. Striated lines of the sarcomere are visible in a T E M at scale 0.5 millimeters. The sarcomere consists of lines of alternating thick (myosin)

filaments and thin (actin) filaments. Z lines attach to thin filaments at the edges of the sarcomere and the thick filaments are anchored at the M line in the center of the sarcomere.

VISUAL SKILLS*Looking at this figure, would you say that there are multiple sarcomeres per myofibril, multiple myofibrils per sarcomere, or both? Explain.*

*For suggested answer, see*Appendix A.**

The myofibrils in muscle fibers are made up of repeating sections called **sarcomeres**, which are the basic contractile units of skeletal muscle. The borders of the sarcomere line up in adjacent myofibrils, forming a pattern of light and dark bands (striations) visible with a light microscope. For this reason, skeletal muscle is also called *striated muscle*. Thin filaments attach at the Z lines at the sarcomere ends, while thick filaments are anchored in the middle of the sarcomere (M line).

In a resting (relaxed) myofibril, thick and thin filaments partially overlap. Near the edge of the sarcomere there are only thin filaments, whereas the zone in the center contains only thick filaments. This partially overlapping arrangement is the key to how the sarcomere, and hence the whole muscle, contracts.

The Sliding-Filament Model of Muscle Contraction

A contracting muscle shortens, but the filaments that bring about contraction stay the same length. To explain this apparent paradox, we'll focus first on a single sarcomere. As shown in **Figure 50.27**, the filaments slide past each other, much like the segments of a telescoping support pole. According to the well-accepted **sliding-filament model**, the thin and thick filaments ratchet past each other, powered by myosin molecules.

Figure 50.27

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The sliding-filament model of muscle contraction.
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Filament length does not change as muscle contracts; instead degree of filament overlap within the sarcomere changes, increasing as the fiber contracts and the filaments slide past each other.

The drawings on the left show that the lengths of the thick (myosin) filaments (purple) and thin (actin) filaments (orange) remain the same as a muscle fiber contracts.

In the sarcomere, thin filaments anchor at the Z lines (at edges of sarcomere) while thick filaments anchor at the M line in the center of the sarcomere. Filament length does not change as the fiber contracts; instead degree of overlap of thick and thin filaments within the sarcomere changes, increasing as the fiber contracts and the filaments slide past each other. In a relaxed muscle, degree of filament overlap is lowest and the

thick filaments do not reach the Z lines (only the thin filaments do), while the thin filaments do not reach the M line. In a contracting muscle, degree of overlap increases as the filaments begin to slide past each other, the thin filaments are closer to the M line, and the thick filaments are closer to the Z lines. In a fully contracted muscle, the thin and thick filaments overlap completely, the thin filaments overlap each other at the M line, and the thick filaments reach the Z lines. Increasing degree of overlap as the filaments slide past each other is visible as darkening lines at scale 0.5 micrometers.

BioFlix® Animation

From Muscle Cells to Movement

image

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image

Figure 50.28 illustrates the cycles of change in the myosin molecule that convert the chemical energy of ATP into the longitudinal sliding of thick and thin filaments.

Figure 50.28

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    Myosin–actin interactions underlying muscle fiber
contraction.
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1. Myosin head in low-energy configuration. 2. ATP hydrolyzed and myosin head in high-energy configuration. 3. Cross-bridge formed. 4. Thin filament slides towards sarcomere center. 5. New ATP molecule binds.

1. On thick filament, myosin head is bound to ATP and is in its low-energy configuration.
2. The myosin head hydrolyzes ATP to ADP and phosphate and is in its high-energy configuration. Myosin binding sites visible on thin filament.
3. The myosin head binds to actin, forming a cross bridge. ADP and phosphate are bound to myosin head.
4. Releasing ADP and inorganic phosphate, myosin returns to its low-energy configuration, sliding the thin filament. The thin filament moves towards the center of the sarcomere.
5. Binding of a new molecule of ATP releases the myosin head from actin, and a new cycle begins.

When ATP binds the myosin head, what prevents the filaments from sliding back into their original positions?

*For suggested answer, see*Appendix A*.*

BioFlix® Animation

Sliding-Filament Model of Muscle Contraction

image

image

As shown in the figure, each myosin molecule has a long “tail” region and a globular “head” region. The tail adheres to the tails of other myosin molecules, binding together the thick filament. The head, jutting to the side, can bind ATP. Hydrolysis of bound ATP converts myosin to a high-energy form that binds to actin, forming a cross-bridge between the myosin and the thin filament. The myosin head then returns to its low-energy form as it helps to pull the thin filament toward the center of the sarcomere. When a new ATP molecule binds to the myosin head, the cross-bridge is disrupted, releasing the myosin head from the actin filament.

Muscle contraction requires repeated cycles of binding and release. During each cycle of each myosin head, the head is freed from a cross-bridge, cleaves the newly bound ATP, and binds again to actin. Because the thin filament moves toward the center of the sarcomere in each cycle, the myosin head now attaches to a binding site farther along the thin filament than in the previous cycle. Each end of a thick filament contains approximately 300 heads, each of which forms and re-forms about five cross-bridges per second, driving the thick and thin filaments past each other.

At rest, most muscle fibers contain only enough ATP for a few contractions. Powering repetitive contractions requires two other compounds found in muscle cells: creatine phosphate and glycogen. The transfer of a phosphate group from creatine phosphate to ADP in an enzyme-catalyzed reaction synthesizes additional ATP. In this way, the resting supply of creatine phosphate can sustain contractions for about 15 seconds. Glycogen is broken down into glucose, which can then be used to replenish ATP stores. During light or moderate muscle activity, this glucose is fully metabolized by aerobic respiration (see Concept 9.1). On average, stored glycogen alone can power sustained contractions for nearly an hour.

Muscle cells also use nutrients in the blood to generate ATP. Glucose liberated from liver glycogen and fatty acids released from fat cells are particularly important during low-intensity, long-duration activities. During intense muscle activity, many glucose molecules undergo glycolysis, but most of the resulting pyruvate molecules are converted into lactate rather than undergoing the remaining steps of aerobic respiration (see

Concept 9.5). This pathway, which does not require either oxygen or mitochondria, is very rapid. However, it generates much less ATP per glucose molecule and can sustain contraction for only a short time.

The Role of Calcium and Regulatory Proteins

Proteins bound to actin play crucial roles in controlling muscle contraction. In a muscle fiber at rest, **tropomyosin**, a regulatory protein, and the **troponin complex**, a set of additional regulatory proteins, are bound to the actin strands of thin filaments. Tropomyosin covers the myosin-binding sites along the thin filament, preventing actin and myosin from interacting **(Figure 50.29a)**.

Figure 50.29

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    The role of regulatory proteins and calcium in muscle fiber
contraction.
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The role of regulatory proteins and calcium in muscle fiber contraction.

Each thin filament consists of two strands of actin and associated regulatory proteins: two long molecules of tropomyosin and multiple copies of the troponin complex.

Parts are as follows.

Part a. Myosin-binding sites blocked by tropomyosin. Tropomyosin and troponin complex bound to actin strands. Calcium ion binding sites in troponin complex are empty and myosin-binding sites are not exposed.

Part b. Myosin-binding sites exposed. Calcium ions bind to troponin complex, exposing myosin-binding sites.

Motor neurons enable actin and myosin to interact by triggering a release of calcium ions (Ca^{2+}) into the cytosol. Once in the cytosol, Ca^{2+} binds to the troponin complex, causing the myosin-binding sites on actin to be exposed **(Figure 50.29b)**. Note that the effect of Ca^{2+} is indirect: Binding to Ca^{2+} causes the troponin complex to change shape, dislodging tropomyosin from the myosin-binding sites.

When the Ca^{2+} concentration rises in the cytosol, the cycle of cross-bridge formation begins, the thin and thick filaments slide past each, and the muscle fiber contracts. When the Ca^{2+} concentration falls, the binding sites are covered, and contraction stops.

Motor neurons cause muscle contraction through a multi-step process triggering the movement of Ca^{2+} into the cytosol of muscle cells. This process is summarized in **Figure 50.30** and detailed step by step in **Figure 50.31**. First, the arrival of an action potential at the synaptic terminal of a motor neuron **1** causes release of the neurotransmitter acetylcholine. Binding of acetylcholine to receptors on the muscle fiber leads to a

depolarization that initiates an action potential. Within the muscle fiber, the action potential spreads deep into the interior, following infoldings of the plasma membrane called **transverse tubules (T)**. **2** These make close contact with the **sarcoplasmic reticulum (SR)**, a specialized endoplasmic reticulum. As the action potential spreads along the T tubules, it triggers changes in the SR, opening Ca2+ channels **3**. Calcium ions stored in the interior of the SR flow through open channels into the cytosol **4** and bind to the troponin complex, **5** initiating the muscle fiber contraction.

Figure 50.30

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    The roles of the sarcoplasmic reticulum and T tubules in
muscle fiber contraction.
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Cutaway view of muscle cell: axon of motor neuron at synaptic terminal; T tubule; sarcoplasmic reticulum (SR) which releases calcium ion; myofibril; plasma membrane of muscle fiber; sarcomere and mitochondria.

The synaptic terminal of a motor neuron releases acetylcholine, which depolarizes the plasma membrane of the muscle fiber. The depolarization causes action potentials (red arrows) to sweep across the muscle fiber and deep into it along the transverse (T) tubules. The action potentials trigger the release of calcium (green dots) from the sarcoplasmic reticulum into the cytosol. Calcium initiates the sliding of filaments by allowing myosin to bind to actin.

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Muscle Structure

image

Play

image

Figure 50.31

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    Summary of contraction in a skeletal muscle fiber.
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Skeletal muscle contraction shown in seven steps.

The steps are as follows.

1.Acetylcholine (A Ch) released at the synaptic terminal diffuses across the synaptic cleft and binds to receptor proteins on the muscle fiber‘s plasma membrane, triggering an action potential in the muscle fiber.

2.The action potential is propagated along the plasma membrane and down T tubules.

3.The action potential triggers C a 2 plus release from the S R.

4.Calcium ions bind to troponin in the thin filament; myosin-binding sites are exposed.

5.Cycles of myosin cross-bridge formation and breakdown, coupled with A T P hydrolysis, slide the thin filament toward the center of the sarcomere.

6.After the action potential ends, cytosolic C a 2 plus is removed by active transport into the S R.

7.Once cytosolic C a 2 plus is removed, tropomyosin again blocks the myosin-binding sites. Contraction ends, and the muscle fiber relaxes.

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Roles of Calcium and ATP in Muscle Contraction

image

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When motor neuron input stops, the filaments slide back to their starting position as the muscle relaxes. Relaxation begins as proteins in the SR pump Ca2+ back into the SR **6** from the cytosol. When the Ca2+ concentration in the cytosol drops to a low level, the regulatory proteins bound to the thin filament shift back to their starting position, **7** once again blocking the myosin-binding sites. At the same time, the Ca2+ pumped from the cytosol accumulates in the SR, providing the stores needed to respond to the next action potential.

Several diseases cause paralysis by interfering with the excitation of skeletal muscle fibers by motor neurons. In amyotrophic lateral sclerosis (ALS), motor neurons in the spinal cord and brainstem degenerate, and muscle fibers atrophy. ALS is progressive and usually fatal within five years after symptoms appear. In myasthenia gravis, a person produces antibodies to the acetylcholine receptors of skeletal muscle. As the disease progresses and the number of receptors decreases, transmission between motor neurons and muscle fibers declines. Myasthenia gravis can generally be controlled with drugs that inhibit acetylcholinesterase or suppress the immune system.

Nervous Control of Muscle Tension

Whereas contraction of a single skeletal muscle fiber is a brief all-or-none twitch, contraction of a whole muscle, such as the biceps in your upper arm, is graded; you can voluntarily alter the extent and strength of its contraction. The nervous system produces graded contractions of whole muscles by varying (1) the number of muscle fibers that contract and (2) the rate at which muscle fibers are stimulated. Let's consider each mechanism in turn.

In vertebrates, each branched motor neuron may form synapses with many muscle fibers, although each fiber is controlled by only one motor neuron. A **motor unit** consists of a single motor neuron and all the muscle fibers it controls **(Figure 50.32)**. When a motor neuron produces an action potential, all the muscle fibers in its motor unit contract as a group. The strength of the resulting contraction depends on how many muscle fibers the motor neuron controls.

Figure 50.32

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Motor units in a vertebrate skeletal muscle.
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Motor units consist of motor neurons and the muscle fibers each neuron controls. Motor neurons connect muscle fibers to the spinal cord. Bundles of muscle fibers make up the muscle.

Each muscle fiber (cell) forms synapses with only one motor neuron, but each motor neuron typically synapses with many muscle fibers. A motor neuron and all the muscle fibers it controls constitute a motor unit.

Motor neuron cell bodies connect to spinal cord. Motor neuron axons connect motor neuron cell bodies to synaptic terminals on muscle fibers. Axons of several neurons are bundled as a nerve. Motor units consist of a motor neuron and all muscle fibers it controls (connects to). Two motor units are depicted, one controlling two muscle fibers and the other controlling three. Bundles of muscle fibers make up a muscle, which is connected to bone by a tendon.

In the whole muscle, there may be hundreds of motor units. As more and more of the corresponding motor neurons are activated, a process called *recruitment*, the force (tension) developed by a muscle progressively increases. Depending on the number of motor neurons your brain recruits and the size of their motor units, you can lift a fork or something much heavier, like a biology textbook. Some muscles, especially those that hold up the body and maintain posture, are almost always partially contracted. In such muscles, the nervous system may alternate activation among the motor units, reducing the length of time any one set of fibers is contracted.

The nervous system regulates muscle contraction not only by controlling which motor units are activated but also by varying the rate of muscle fiber stimulation. A single action potential produces a twitch lasting about

100 milliseconds or less. If a second action potential arrives before the muscle fiber has completely relaxed, the two twitches add together, resulting in greater tension **(Figure 50.33)**. Further summation occurs as the rate of stimulation increases. When the rate is so high that the muscle fiber cannot relax at all between stimuli, the twitches fuse into one smooth, sustained contraction called **tetanus**. (Note that tetanus is also the name of a disease of uncontrolled muscle contraction caused by a bacterial toxin.)

Figure 50.33

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Summation of twitches.
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A graph of tension versus time for an action potential, a pair of action potentials, and series of action potentials at high frequency. Tension is highest with increasing summed action potentials.

This graph illustrates how the number of action potentials during a short period of time influences the tension developed in a muscle fiber.

One action potential produces a single small hump representing a single twitch. The summation of a pair of action potentials (two twitches) creates a taller hump. A series of action potentials at high frequency creates a much higher hump, with tetanus achieved as the peak flattens at high tension.

How could the nervous system cause a skeletal muscle to produce the most forceful contraction it is capable of?

*For suggested answer, see*Appendix A.**

Types of Skeletal Muscle Fibers

Our discussion to this point has focused on the general properties of vertebrate skeletal muscles. There are, however, several distinct types of skeletal muscle fibers, each of which is adapted to a particular set of functions. We typically classify these varied fiber types both by the source of ATP used to power their activity and by the speed of their contraction **(Table 50.1)**.

Table 50.1

Types of Skeletal Muscle Fibers

A table titled Table 50.1 Types of skeletal muscle fibers illustrates slow twitch and fast twitch process.

Table 50.1 Types of Skeletal Muscle Fibers

Oxidative
Contraction speed
Major A T P source
Rate of fatigue
Mitochondria
Myoglobin content

Oxidative and Glycolytic Fibers

Fibers that rely mostly on aerobic respiration are called oxidative fibers. Such fibers are specialized in ways that enable them to make use of a steady energy supply: They have many mitochondria, a rich blood supply, and a large amount of an oxygen-storing protein called **myoglobin**. A brownish red pigment, myoglobin binds oxygen more tightly than does hemoglobin, enabling oxidative fibers to extract oxygen from the blood efficiently. In contrast, glycolytic fibers have a larger diameter and less myoglobin. Also, glycolytic fibers use glycolysis as their primary source of ATP and fatigue more readily than oxidative fibers. These two fiber types are readily apparent in the muscle of poultry and fish: The dark meat (red muscle) is made up of oxidative fibers rich in myoglobin, and the light meat (white muscle) is composed of glycolytic fibers.

Fast-Twitch and Slow-Twitch Fibers

Muscle fibers vary in the speed with which they contract: **Fast-twitch fibers** develop tension two to three times faster than **slow-twitch fibers**. Fast fibers enable brief, rapid, powerful contractions. Compared with a fast fiber, a slow fiber has less sarcoplasmic reticulum and pumps Ca^{2+} more slowly. Because Ca^{2+} remains in the cytosol longer, a muscle twitch in a slow fiber lasts about five times as long as one in a fast fiber.

The difference in contraction speed between slow-twitch and fast-twitch fibers mainly reflects the rate at which their myosin heads hydrolyze ATP. However, there isn't a one-to-one relationship between contraction speed and ATP source. Whereas all slow-twitch fibers are oxidative, fast-twitch fibers can be either glycolytic or oxidative.

Most human skeletal muscles contain both fast-twitch and slow-twitch fibers, although the muscles of the eye and hand are exclusively fast-twitch. In a muscle that has a mixture of fast and slow fibers, the relative proportions of each are genetically determined. However, if such a muscle is used repeatedly for activities requiring high endurance, some fast glycolytic fibers can develop into fast oxidative fibers. Because fast

oxidative fibers fatigue more slowly than fast glycolytic fibers, the result will be a muscle that is more resistant to fatigue.

Some vertebrates have skeletal muscle fibers that twitch at rates far faster than any human muscle. For example, superfast muscles produce a rattlesnake's rattle and a dove's coo. Even faster are the muscles surrounding the gas-filled swim bladder of the male oyster toadfish **(Figure 50.34)**. In producing its "boat whistle" mating call, the toadfish can contract and relax these muscles more than 200 times per second!

Figure 50.34

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Specialization of skeletal muscles.
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Male toadfish (Opsanus tau).

The male oyster toadfish (Opsanus tau) uses superfast muscles to produce its mating call.

Other Types of Muscle

Although all muscles share the same fundamental mechanism of contraction—actin and myosin filaments sliding past each other—there are many different types of muscle. Vertebrates, for example, have cardiac muscle and smooth muscle in addition to skeletal muscle (see Figure 40.5).

Vertebrate **cardiac muscle** is found only in the heart and, like skeletal muscle, is striated. Unlike the cells of skeletal muscle fibers, some cardiac muscle cells can initiate rhythmic depolarization and contraction without nervous system input. Normally, however, cells in one part of the heart act as a pacemaker to initiate contraction. Signals from the pacemaker spread throughout the heart because specialized regions called *intercalated disks* electrically couple each cardiac muscle cell to the adjacent cells. It is this coupling that enables action potentials generated in one part of the heart to trigger contraction throughout the organ. Although these action potentials last up to 20 times longer than those of skeletal muscle fibers, a long refractory period prevents summation and tetanus.

Smooth muscle in vertebrates is found in the walls of hollow organs, such as vessels and tracts of the circulatory, digestive, and reproductive systems. In these locations it controls blood flow in the arteries, moves food through the digestive system, carries out uterine contractions during labor, and assists in regulating the temperature of the testicles. Smooth muscle is also found in the eye, where its action controls focusing and pupil diameter.

Smooth muscle cells lack striations because their actin and myosin filaments are not regularly arrayed along the length of the cell. Instead, the thick filaments are scattered throughout the cytoplasm, and the thin

filaments are attached to structures called dense bodies, some of which are tethered to the plasma membrane. There is less myosin than in striated muscle fibers, and the myosin is not associated with specific actin strands. Some smooth muscle cells contract only when stimulated by neurons of the autonomic nervous system. Others are electrically coupled to one another and can generate action potentials without input from neurons. Smooth muscles contract and relax more slowly than striated muscles.

Although Ca^{2+} regulates smooth muscle contraction, smooth muscle cells have no troponin complex or T tubules, and their sarcoplasmic reticulum is not well developed. During an action potential, Ca^{2+} enters the cytosol mainly through the plasma membrane. Calcium ions cause contraction by binding to the protein calmodulin, which activates an enzyme that phosphorylates the myosin head, enabling cross-bridge activity.

Invertebrates have muscle cells similar to vertebrate skeletal and smooth muscle cells; in fact, arthropod skeletal muscles are nearly identical to those of vertebrates. However, because the flight muscles of insects contract in response to stretching, the wings of some insects can actually beat up and down faster than action potentials can arrive from the central nervous system. Another interesting evolutionary adaptation has been discovered in the muscles that hold a clam's shell closed. Modification of certain proteins in these muscles allows them to remain contracted for as long as a month with only a low rate of energy consumption.